

Internship Report

Submitted

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B.Tech CSE



By

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Batch: **2CSE08**

2026-27

CANDIDATE'S DECLARATION

I, Lakshya Saxena do hereby declare that the internship report titled “**Analysis of Youtube Ad Conversion Performance**” has been completed by me to fulfil the requirement for the award of the degree of Bachelor of Technology in CSE. All the references have been quoted and I have not taken as such any material from any other source. I affirm to you that this report is my own and has not been submitted to any other institute for any degree/diploma requirement and further I shall be solely responsible for any kind of copyright violation in this regard.

Signature

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ACKNOWLEDGEMENT

I am using this opportunity to express my gratitude to everyone who supported me throughout the Internship Program. I am thankful for their aspiring guidance, invaluable constructive criticism and friendly advice during this work. I am sincerely grateful to them for sharing their truthful and illuminating views on a number of issues related to this work.

I express my gratitude to my parents for their blessings.

Signature

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INTRODUCTION

In today's digital marketing landscape, understanding the effectiveness of advertising campaigns is crucial for optimizing return on investment (ROI). Youtube Ads have become one of the most powerful tools for businesses to reach their target audiences, offering granular targeting capabilities based on demographics, interests, and behaviour. However, the vast amount of data generated by these campaigns presents both opportunities and challenges for marketers and data analysts.

This project focuses on analysing a dataset of Youtube ad campaigns to understand the factors that influence conversion rates, optimize ad spend, and provide actionable insights for improving campaign performance. The dataset contains information on various ad campaigns, including impressions, clicks, spending, and conversions across different age groups, genders, and interest categories.

As a Data Analyst Intern at Flairmark Entertainments, I was tasked with analysing this dataset to derive meaningful insights that could help the marketing team make data-driven decisions. This report documents the entire process, from data exploration to analysis and recommendations.

The analysis is built on the understanding that successful ad campaigns are not just about reaching a large audience but about reaching the *right* audience with the *right* message at the *right* time. By identifying patterns in conversion data, we can uncover which segments of the audience are most likely to convert and which ad creatives or targeting strategies are most effective.

ORGANISATION PROFILE

Flairmark Entertainments is a dynamic and innovative entertainment company dedicated to creating memorable experiences through various forms of media and entertainment. The company operates in the digital entertainment space, leveraging social media platforms to promote events, content, and merchandise to a diverse audience.

Core Business Areas:

- Event promotion and management
- Digital content creation and distribution
- Brand partnerships and influencer marketing
- Social media advertising and campaign management

Vision: To be a leading force in the entertainment industry by creating engaging content and experiences that resonate with audiences worldwide.

Mission: To deliver exceptional entertainment experiences through innovative marketing strategies, creative content, and data-driven decision making.

Internship Role: Data Analyst Intern

During my internship at Flairmark Entertainments, I worked with the marketing team to analyse advertising campaign data, develop dashboards and reports, and provide insights to optimize marketing spend. My responsibilities included:

- Collecting and cleaning advertising data from various sources
- Performing exploratory data analysis to identify trends and patterns
- Building visualization dashboards to track key performance indicators
- Generating reports and presentations for stakeholders

PROBLEM STATEMENT

Flairmark Entertainments runs multiple Youtube ad campaigns to promote their events and content. However, the marketing team faces several challenges:

1. **High Advertising Costs:** The company spends significant amounts on Facebook Ads but lacks clarity on which campaigns and demographics are generating the highest ROI.
2. **Low Conversion Rates:** Despite high impressions and clicks, the conversion rates are not meeting expectations, suggesting that the targeting or ad content may not be effectively reaching the right audience.
3. **Inefficient Budget Allocation:** Without a clear understanding of which audience segments and campaigns are most effective, the marketing team struggles to allocate budgets optimally.
4. **Lack of Data-Driven Insights:** The team needs actionable insights to improve campaign performance, including understanding which factors (age, gender, interest, ad creative) most strongly influence conversions.

Key Questions to Answer:

- Which age groups and genders have the highest conversion rates?
- What is the relationship between ad spend, impressions, clicks, and conversions?
- Which interests are associated with higher conversion rates?
- How do different campaigns perform in terms of ROI?
- What are the recommendations for optimizing future campaigns?

PROJECT OBJECTIVES

The primary objectives of this project are:

Objective 1: Data Exploration and Preparation

- Understand the structure and content of the dataset
- Clean the data by handling missing values and outliers
- Create new derived features (e.g., CTR, Conversion Rate, ROI)

Objective 2: Performance Analysis

- Analyse the performance of ad campaigns across different dimensions (age, gender, interest, campaign ID)
- Calculate and visualize key metrics such as Click-Through Rate (CTR), Conversion Rate (CVR), and Return on Ad Spend (ROAS)

Objective 3: Identifying High-Performing Segments

- Identify demographic segments (age-gender combinations) with the highest conversion rates
- Analyse which interest categories are most effective for driving conversions
- Determine the optimal ad spend range for achieving high ROI

Objective 4: Insights and Recommendations

- Provide actionable insights based on the analysis
- Recommend budget allocation strategies for future campaigns
- Suggest improvements in targeting and ad design based on data patterns

Objective 5: Documentation and Presentation

- Document the analysis process and findings
- Create visualizations and dashboards to present insights to stakeholders

SCOPE OF THE PROJECT

In-Scope:

- Analysis of the provided Youtube ad dataset (ad_conversion.csv)
- Study of demographic factors (age, gender, interest) on campaign performance
- Evaluation of key performance metrics (impressions, clicks, spend, conversions)
- Recommendations for improving future campaign performance

Out-of-Scope:

- Analysis of ad creative content (images, videos, copy)
- Integration with external data sources
- Real-time data analysis or dashboard
- Implementation of advertising campaigns

Limitations:

- The dataset may not be representative of all Youtube ad campaigns
- Limited information about external factors that could influence campaign performance (seasonality, competition, etc.)
- The analysis is based on historical data and may not perfectly predict future performance

TECHNOLOGIES AND TOOLS USED

Programming Language:

- **Python 3.8+** - Primary language for data analysis and scripting

Libraries and Frameworks:

Library	Purpose
Pandas	Data manipulation and analysis
NumPy	Numerical computing and mathematical operations
Matplotlib	Data visualization and plotting
Seaborn	Statistical data visualization
Scikit-learn	Machine learning
Jupyter Notebook	Interactive development environment

Tools:

- **Jupyter Notebook/Google Colab** - For interactive data analysis and visualization
- **Excel** - For additional data processing and validation
- **PowerPoint** - For creating presentations of findings

Statistical Techniques:

- Descriptive statistics (mean, median, standard deviation)
- Correlation analysis
- Group-wise aggregation and comparison
- Conversion rate analysis

SYSTEM ARCHITECTURE

DATA ACQUISITION

ad_conversion.csv

DATA PREPROCESSING

Data loading and initial exploration
Handling missing values
Data type conversion
Feature engineering

EXPLORATORY ANALYSIS

Descriptive statistics
Distribution analysis
Correlation analysis
Group-wise aggregation

PERFORMANCE ANALYSIS

Campaign performance by demographics
Interest-based analysis
Spend efficiency analysis

VISUALIZATION & INSIGHTS

Charts and graphs
Dashboard design
Key findings and recommendations

METHODOLOGY

The methodology for this project follows a systematic approach to data analysis, ensuring that the findings are robust, reliable, and actionable. The methodology is outlined in the following phases:

Phase 1: Data Understanding and Preparation

1.1 Data Loading and Initial Exploration

- Load the ad_conversion.csv file using Pandas
- Examine the structure of the dataset (number of rows, columns, data types)
- Display first few rows to understand the data format

1.2 Data Cleaning

- Check for missing values and decide on handling strategies (drop, impute, or mark)
- Remove duplicate rows if any
- Standardize data formats where necessary

1.3 Feature Engineering

- Calculate derived metrics:
 - **Click-Through Rate (CTR):** $(\text{Clicks} / \text{Impressions}) \times 100\%$
 - **Conversion Rate (CVR):** $(\text{Approved_Conversion} / \text{Total_Conversion}) \times 100\%$
 - **Cost Per Conversion:** $\text{Spent} / \text{Total_Conversion}$
 - **Return on Ad Spend (ROAS):** $\text{Revenue} / \text{Spent}$ (assuming value per conversion)
 - **Cost Per Click (CPC):** $\text{Spent} / \text{Clicks}$

Phase 2: Exploratory Data Analysis (EDA)

2.1 Univariate Analysis

- Distribution of numerical variables (Impressions, Clicks, Spent, Conversions)
- Distribution of categorical variables (Age, Gender, Campaign ID)

2.2 Bivariate Analysis

- Relationship between spend and impressions
- Relationship between impressions and conversions
- Relationship between clicks and conversions

2.3 Multivariate Analysis

- Performance across different age groups
- Performance across genders
- Impact of interest categories on conversions

Phase 3: Performance Analysis

3.1 Campaign Performance Metrics

- Calculate overall CTR, CVR, and ROAS
- Identify top-performing campaigns

3.2 Demographic Analysis

- Compare conversion rates across age groups (30-34, 35-39, 40-44, 45-49)
- Compare performance between Male and Female audiences

3.3 Interest-Based Analysis

- Identify interest categories with highest conversion rates
- Analyse spending patterns across interest categories

3.4 Campaign Efficiency Analysis

- Analyse relationship between spend and conversions
- Identify optimal spend ranges for achieving high conversion

Phase 4: Visualization and Reporting

4.1 Dashboard Creation

- Create visualizations to represent key findings
- Use appropriate chart types:
 - Bar charts for comparing categories
 - Line charts for trends
 - Scatter plots for relationships
 - Histograms for distributions

4.2 Insight Generation

- Summarize key findings
- Identify actionable insights for the marketing team

4.3 Recommendations

- Provide specific recommendations for:
 - Budget allocation
 - Audience targeting
 - Campaign optimization

Phase 5: Documentation

5.1 Code Documentation

- Comment code appropriately
- Document data processing steps

5.2 Report Writing

- Document the analysis process and findings
- Create a comprehensive report for stakeholders

EXPECTED OUTCOMES

Based on the analysis of the Facebook ad conversion dataset, the following outcomes are expected:

Outcome 1: Identification of Best Performing Demographics

The analysis is expected to reveal which age, and gender combinations yield the highest conversion rates. For example, the data shows significant spending on campaigns targeting women aged 45-49, which may indicate that this demographic is particularly valuable for the company's offerings.

Key Observations from Data:

- There is a strong presence of campaigns targeting females in the 30-34 and 45-49 age groups
- Male audiences in the 30-34 and 35-39 age groups are also heavily targeted
- Interest categories vary significantly across different campaigns

Outcome 2: Insights on Campaign Efficiency

The analysis will reveal which campaigns are most efficient in terms of converting impressions and spend into conversions. This will help in understanding:

- Which campaign IDs consistently perform well
- How the xyz_campaign_id and fb_campaign_id impact performance
- The relationship between audience targeting and conversion rates

Outcome 3: ROI Optimization Recommendations

The project is expected to produce specific recommendations for:

- **Budget Allocation:** How to distribute budgets across campaigns, age groups, and genders
- **Targeting Strategy:** Which audience segments to prioritize for future campaigns
- **Ad Design:** Insights into what kinds of ads (based on interest categories) resonate most with different audiences
- **Campaign Management:** Suggestions on when to increase or decrease spending based on performance patterns

Outcome 4: Interactive Dashboard

A set of visualizations will be created to allow the marketing team to:

- Monitor campaign performance in real-time
- Compare metrics across different segments
- Identify underperforming campaigns quickly

Outcome 5: Actionable Recommendations

The final deliverable will be a set of actionable recommendations, including:

1. **Audience Recommendations:** "Focus on women aged 45-49 with interests in [specific categories]"
2. **Budget Recommendations:** "Allocate 40% of the budget to campaign X based on its superior ROI"
3. **Timing Recommendations:** "Increase spending during [specific times] when conversion rates are highest"
4. **A/B Testing Suggestions:** "Test different ad creatives targeting [segment] to further improve conversion rates"

CERTIFICATE



CERTIFICATE ID:
FM-INT-2026-125



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OF INTERNSHIP

THIS IS TO CERTIFY THAT

Lakshay Saxena

has successfully completed the Internship Program at **Flairmark Entertainments**
in the position of **Data Analyst Intern** during the period of
13 May 2026 to 12 August 2026 (3 Months).
During this internship, they have demonstrated dedication, hard work,
a positive attitude, and a strong willingness to learn.
We wish them all the best for their future endeavors.

 INTERNSHIP DURATION 13 May 2026 to 12 August 2026 (3 Months)	 DEPARTMENT Data Analytics	 LOCATION Chandigarh	 ROLE Data Analyst Intern
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ANOOP KUMAR
Founder & CEO
Flairmark Entertainments

STIPEND
₹5,000/-
PER MONTH

DATE OF ISSUE
10 May 2026

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